

Mathematical Methods for Finance

Module 1 – Completeness, Compactness and Normed Spaces

Lesson 4: Completeness, Baire, compactness and normed spaces

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3-hour lesson structure: intuition, theory, examples, exercises, and exam-style practice

Learning outcomes for Lesson 4

By the end of this lesson, students should be able to:

- ① use completeness in fixed point and approximation arguments;
- ② state the Baire Category Theorem and interpret its meaning;
- ③ recognize compactness through sequential compactness and Heine–Borel;
- ④ distinguish finite-dimensional and infinite-dimensional phenomena;
- ⑤ verify norms and use Holder and Minkowski inequalities;
- ⑥ work with basic sequence and function spaces.

How today is organised

14:30–15:20

Concepts and intuition

Completeness, contractions and Baire.

15:35–16:25

Theory and examples

Compactness and Heine–Borel.

16:40–17:30

Exercises

Normed spaces, inequalities and concrete spaces.

Intuition: Learning rule

Each new concept follows the same route: picture, definition, worked example, counterexample, exercise.

Module 1

Block 1: completeness, contractions and Baire

Recap: completeness

Definition: Complete metric space

A metric space is complete if every Cauchy sequence has a limit inside the space.

Example: Complete

$\mathbb{R}$, $\mathbb{R}^n$, closed subsets of complete spaces,
 $C([a, b])$ with $\|\cdot\|_\infty$.

Counterexample: Not complete

$\mathbb{Q}$, $(0, 1)$, many spaces with the wrong norm
or missing limits.

Today

We use completeness to guarantee the existence of limits produced by iterative methods.

Contractions

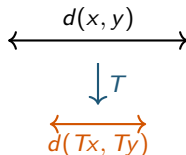
Definition: Contraction

Let (X, d) be a metric space. A map $T : X \rightarrow X$ is a contraction if there exists $0 < q < 1$ such that

$$d(Tx, Ty) \leq q d(x, y) \quad \forall x, y \in X.$$

Intuition: Meaning

Applying T pulls points closer together by a uniform factor.



Banach fixed point theorem

Theorem: Contraction theorem

Let (X, d) be complete and let $T : X \rightarrow X$ be a contraction. Then there exists a unique $x^* \in X$ such that

$$Tx^* = x^*.$$

Moreover, for any starting point x_0 , the iteration $x_{n+1} = Tx_n$ converges to x^* .

Exam message

The hypotheses are not decorative: completeness and contraction are essential.

Proof idea: contraction theorem

Proof idea: Three steps

- 1 Define $x_{n+1} = Tx_n$.
- 2 Show (x_n) is Cauchy using repeated contraction:

$$d(x_{n+1}, x_n) \leq q^n d(x_1, x_0).$$

- 3 Use completeness to obtain a limit x^* , then continuity of T gives $Tx^* = x^*$.

Uniqueness follows because if $Tx = x$ and $Ty = y$, then

$$d(x, y) = d(Tx, Ty) \leq qd(x, y),$$

so $d(x, y) = 0$.

Worked example: linear contraction

Worked example: $T(x) = \frac{1}{2}x + 1$ on $\mathbb{R}$

For all $x, y \in \mathbb{R}$,

$$|T(x) - T(y)| = \frac{1}{2} |x - y|.$$

Thus T is a contraction with $q = 1/2$. The fixed point satisfies

$$x = \frac{1}{2}x + 1 \quad \Rightarrow \quad x = 2.$$

Exercise: Iteration ★★

Starting from $x_0 = 0$, compute x_1, x_2, x_3 and observe convergence to 2.

Why completeness matters in fixed points

Counterexample: Missing the limit

Let $X = (0, 1)$ and $T(x) = x/2$. Then $T : X \rightarrow X$ is a contraction, but the fixed point equation gives $x = 0$, which is not in X .

Conclusion

The space must contain the limit produced by the iteration.

Baire category theorem: intuition

Intuition: The message

A complete metric space cannot be built by stacking countably many closed sets that are all too thin.

Definition: Nowhere dense

A set A is nowhere dense if the interior of its closure is empty:

$$(\overline{A})^\circ = \emptyset.$$

Example: Mental image

Single points in $\mathbb{R}$ are nowhere dense: no interval is contained in a single point.

Baire category theorem

Theorem: BCT

If (X, d) is a complete metric space, then X cannot be written as a countable union of nowhere dense closed sets. Equivalently, a countable intersection of open dense sets is dense.

Course-level interpretation

Completeness prevents the space from being too small. This theorem is a background reason why complete spaces are structurally robust.

Baire theorem: what to remember

Theorem: Key hypotheses

- complete metric space;
- countably many sets;
- empty interior / nowhere dense sets.

Counterexample: Common mistake

Baire is not a statement about finite unions, and it is not true in arbitrary incomplete spaces.

Exercise: Conceptual check ★★

Why is completeness the natural hypothesis in Baire's theorem?

Block 1 mini-workshop

Exercise: Exercise A ★★

Show that $T(x) = \cos(x)/3$ is a contraction on $\mathbb{R}$.

Exercise: Exercise B ★★★

Explain why $T(x) = x/2$ on $(0, 1)$ does not contradict the contraction theorem.

Exercise: Exercise C ★★

Give an example of a nowhere dense subset of $\mathbb{R}$.

Module 1

Block 2: compactness and Heine–Borel

Compactness: first idea

Intuition: Compact means no escape

A compact set is a set where every sequence has a convergent subsequence with limit still in the set.

Definition: Sequential compactness

A metric space K is sequentially compact if every sequence in K has a subsequence converging to a point of K .

Example: Interval

In $[0, 1]$, a sequence can oscillate or move around, but a subsequence must settle somewhere in $[0, 1]$.

Visual: extracting a subsequence



Interpretation

Compactness turns arbitrary sequences into convergent subsequences.

Heine–Borel theorem

Theorem: Heine–Borel in $\mathbb{R}^n$

A subset $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Example: Examples

- $[0, 1]$ is compact in $\mathbb{R}$.
- $(0, 1)$ is bounded but not compact.
- $[0, +\infty)$ is closed but not compact.
- Closed balls in $\mathbb{R}^n$ are compact.

Why closed and bounded are both needed

Counterexample: Bounded but not closed

In $(0, 1)$, the sequence $x_n = 1/n$ has no subsequence converging to a point of $(0, 1)$.

Counterexample: Closed but not bounded

In $[0, +\infty)$, the sequence $x_n = n$ has no convergent subsequence in $\mathbb{R}$.

Exam habit

When asked about compactness in $\mathbb{R}^n$, check closedness and boundedness separately.

Continuous images of compact sets

Theorem: Compactness preserved by continuous maps

If K is compact and $f : K \rightarrow Y$ is continuous, then $f(K)$ is compact.

Theorem: Extreme value theorem

If $K \subset \mathbb{R}^n$ is compact and $f : K \rightarrow \mathbb{R}$ is continuous, then f attains a maximum and a minimum on K .

Exercise: Example ★★

Does $f(x) = 1/x$ attain a maximum on $(0, 1)$? Which hypothesis fails?

Solution: missing compactness

Proof idea: Analysis

The function $f(x) = 1/x$ is continuous on $(0, 1)$, but $(0, 1)$ is not compact. As $x \rightarrow 0^+$, $1/x \rightarrow +\infty$, so no maximum exists.

Lesson

Continuity alone does not guarantee extrema. Compactness is the missing ingredient.

Finite versus infinite-dimensional compactness

Counterexample: A major warning

In infinite-dimensional normed spaces, closed and bounded sets need not be compact.

Example: Sequence space idea

In an infinite-dimensional space, one can build sequences of points that stay bounded but never have a convergent subsequence.

Intuition: Mental image

Finite-dimensional bounded sets have only finitely many independent directions. Infinite-dimensional spaces have infinitely many directions of escape.

Example: unit vectors in ℓ^2

Example: No convergent subsequence

Let $e_n = (0, \dots, 0, 1, 0, \dots)$ be the sequence with 1 in the n -th position. In ℓ^2 ,

$$\|e_n\| = 1,$$

so the sequence is bounded. But for $m \neq n$,

$$\|e_n - e_m\|_2 = \sqrt{2}.$$

Thus no subsequence can be Cauchy, hence no subsequence can converge.

Conclusion

The closed unit ball of ℓ^2 is not compact.

Compactness mini-workshop

Exercise: Exercise A ★

Decide whether $[2, 5]$ is compact in $\mathbb{R}$.

Exercise: Exercise B ★★

Decide whether $\{x \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}$ is compact.

Exercise: Exercise C ★★★

Explain why the sequence (e_n) in ℓ^2 has no convergent subsequence.

Module 1

Block 3: normed spaces, inequalities and examples

Vector spaces: quick recall

Definition: Vector space

A vector space is a set where vectors can be added and multiplied by scalars, satisfying the usual algebraic rules.

Example: Examples

- $\mathbb{R}^n$;
- polynomials of degree at most m ;
- sequences;
- functions on an interval;
- continuous functions $C([a, b])$.

Why this matters

Linear combinations are the basic operation behind approximation.

Normed spaces

Definition: Norm

A norm on a vector space V is a function $\|\cdot\| : V \rightarrow [0, +\infty)$ such that

$$\|x\| = 0 \iff x = 0, \quad \|\lambda x\| = |\lambda| \|x\|,$$

$$\|x + y\| \leq \|x\| + \|y\|.$$

Intuition: Meaning

A norm measures the size of a vector. It automatically generates a metric:

$$d(x, y) = \|x - y\|.$$

Examples of norms

Example: Norms on $\mathbb{R}^n$

For $x = (x_1, \dots, x_n)$,

$$\|x\|_1 = \sum_{i=1}^n |x_i|, \quad \|x\|_2 = \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2},$$
$$\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

Exercise: Compute ★

For $x = (2, -1, 3)$, compute $\|x\|_1$, $\|x\|_2$ and $\|x\|_\infty$.

Solution: three norms

Worked example: $x = (2, -1, 3)$

$$\|x\|_1 = 2 + 1 + 3 = 6.$$

$$\|x\|_2 = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}.$$

$$\|x\|_\infty = \max\{2, 1, 3\} = 3.$$

Interpretation

Different norms measure size differently, but in finite dimensions they generate the same notion of convergence.

Holder and Minkowski inequalities

Theorem: Holder inequality

If $1 \leq p, q \leq \infty$ and $1/p + 1/q = 1$, then

$$\sum_i |x_i y_i| \leq \|x\|_p \|y\|_q.$$

Theorem: Minkowski inequality

For $p \geq 1$,

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p.$$

Exam message

Holder controls products. Minkowski gives the triangle inequality for L^p and ℓ^p norms.

Cauchy–Schwarz as a special case

Theorem: Cauchy–Schwarz

For $p = q = 2$, Holder becomes

$$\left| \sum_i x_i y_i \right| \leq \left(\sum_i x_i^2 \right)^{1/2} \left(\sum_i y_i^2 \right)^{1/2}.$$

Worked example: Numerical check

Let $x = (1, 2)$ and $y = (3, -1)$. Then

$$|x \cdot y| = |3 - 2| = 1,$$

while $\|x\|_2 \|y\|_2 = \sqrt{5}\sqrt{10} = \sqrt{50}$.

Sequence spaces

Definition: ℓ^p spaces

For $1 \leq p < \infty$,

$$\ell^p = \left\{ x = (x_i) : \sum_{i=1}^{\infty} |x_i|^p < \infty \right\}, \quad \|x\|_p = \left(\sum_i |x_i|^p \right)^{1/p}.$$

Also,

$$\ell^\infty = \{x = (x_i) : \sup_i |x_i| < \infty\}.$$

Exercise: Membership ★★

For $x_i = 1/i$, decide whether $x \in \ell^1$, ℓ^2 and ℓ^∞ .

Solution: sequence membership

Worked example: $x_i = 1/i$

- $x \notin \ell^1$ because $\sum_{i=1}^{\infty} 1/i$ diverges.
- $x \in \ell^2$ because $\sum_{i=1}^{\infty} 1/i^2$ converges.
- $x \in \ell^{\infty}$ because $\sup_i 1/i = 1$.

Lesson

Membership depends on the exponent and on the tail of the sequence.

Function spaces

Definition: Common function spaces

- $C([a, b])$: continuous functions.
- $C^1([a, b])$: continuously differentiable functions.
- $C^k([a, b])$: functions with continuous derivatives up to order k .
- C^∞ : smooth functions.

Example: Two norms

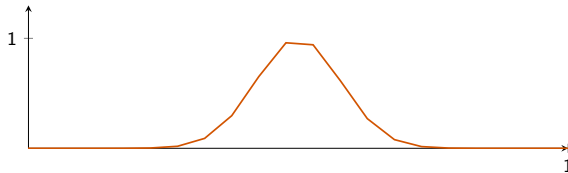
On $C([a, b])$,

$$\|f\|_\infty = \sup_{t \in [a, b]} |f(t)|, \quad \|f\|_1 = \int_a^b |f(t)| dt.$$

A normed-space counterexample

Counterexample: Integral norm can miss pointwise size

On $C([0, 1])$, a function can have small area but large maximum if it is concentrated near a small interval.



Message

The choice of norm determines which differences are considered important.

Normed-space mini-workshop

Exercise: Exercise A ★

Compute $\|(1, -2, 2)\|_1$, $\|(1, -2, 2)\|_2$, $\|(1, -2, 2)\|_\infty$.

Exercise: Exercise B ★★

Use Cauchy–Schwarz to bound $|x_1 + 2x_2 + 3x_3|$ in terms of $\|x\|_2$.

Exercise: Exercise C ★★★

Show that $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ in $\mathbb{R}^n$.

End-of-lesson checklist

Theorem: You should know

- contraction theorem;
- Baire theorem meaning;
- compactness and Heine–Borel;
- finite vs infinite-dimensional compactness;
- norms and norm-generated metrics;
- Holder and Minkowski.

Counterexample: You should avoid

- applying Heine–Borel outside $\mathbb{R}^n$;
- forgetting completeness in fixed point arguments;
- treating all norms as identical in infinite dimensions.

After-class work

Theorem: Core exercises

Exercise pack: 019, 020, 025, 036.

Example: Recommended

Exercise pack: 021, 027, 040, 043.

Advanced

Exercise pack: 031, 035, 058, 060.

Exercise: Study method

For each exercise: write the relevant definition, check all hypotheses, then compute or build a counterexample.

Bridge to Lesson 5

Next question

Once we have normed spaces, the next step is to study complete normed spaces, inner products, orthogonality, projections and linear operators.

Intuition: Guiding image

Hilbert spaces are spaces where infinite-dimensional geometry still resembles Euclidean geometry.